ARUN KUMAR A R

Unity Developer — AR/VR

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SUMMARY

I am Arun Kumar A R ,Unity3D/C# developer with 5+ years of commercial experience, with the bulk of that work concentrated on shipping immersive AR/VR applications on Meta Quest 2 for enterprise clients - Honeywell, MetaGP, GenRobotics, and clinical/education VR platforms. Strong grounding in OOP, SOLID principles, and Unity-native design pattern implementation (Singleton, Observer, State Machine, Object Pool, Factory), with hands-on experience across URP/HDRP pipelines, Addressables, performance profiling, and TCP/IP/mirror networking built from scratch. Comfortable working across the full prototype-to-production cycle under tight deadlines.

TECHNICAL SKILLS

- Core: C# (expert), OOP, SOLID principles, Design Patterns (Singleton, Observer, State Machine, Object Pool, Factory), C++ (learning)
- Unity Engine: Unity 6, Unity 2019/2021/2022 LTS, URP/HDRP, Shader Graph, Addressable Asset System, JSON serialization, MVC
- AR/VR: Meta Quest 2 (Oculus), first-person VR navigation, immersive UI/UX, physics-based interaction design, industrial AR training simulation
- Computer Vision: OpenCV in Unity, Tesseract OCR in Unity
- Networking & Multiplayer: TCP/IP socket programming (built from scratch), Mirror networking, REST APIs, JWT authentication, AWS S3; learning Photon, Netick 2
- Debugging & Optimization: Unity Profiler, memory management, frame rate optimization (sustained 72+ FPS on Quest 2), draw call reduction, LOD systems
- Platforms: Android, iOS, PC, WebGL, VR/XR, cross-platform deployment pipelines
- Tools: Git/GitHub, CI/CD (basic), GitHub Copilot, Cursor, Visual Studio

PROFESSIONAL EXPERIENCE

Unity Developer — Game Development & VR/XR — Tiltlabs Consultancy Services *Technopark, Trivandrum | Feb 2022 – Present*

- Developed MetaGP Garage VR from prototype to production: first-person VR navigation, JWT auth, Addressable Asset System for dynamic loading — sustained 72+ FPS on Meta Quest 2 throughout all build iterations.
- Profiled and debugged the Honeywell AR Training application (petroleum plant simulation), resolving performance bottlenecks and improving training efficiency by 40%+.
- Architected the GenRobotics VR experience with real-time TCP socket-based video streaming on Oculus Quest — networking built end-to-end from scratch.
- Delivered Serenity PathWays VR, a clinical cognitive training app: physics-based puzzles, progressive difficulty scaling, and motor-skill interactions, scoped for clinical use.
- Built KidZania TVS Bike Studio: drag-and-drop customization, automated video recording, QR code generation, and AWS S3 integration, supporting 100+ daily sessions post-launch.
- Built KidZania Mahindra Lifespaces kiosk: draggable building mechanics, dynamic scoring, and certificate generation, iterated from prototype to daily-use production deployment.
- Built functional UI systems across multiple projects; instrumented player/user behavior tracking to inform iteration decisions.

- Used AI coding assistants (GitHub Copilot, Cursor) to accelerate feature decomposition and prototype scaffolding.

Unity Developer — 3D, VR & Game Development — Embright Infotech Pvt Ltd *Kerala, India | Mar 2021 – Feb 2022*

- Developed a VR laser and light reflection simulation on Oculus Quest using Unity Physics, from prototype through production.
- Created interactive learning modules for Auticare VR, an ASD educational platform, contributing to feature design and implementation from prototype through release.
- Built an Alphabet Learning mobile app featuring drag-and-drop gameplay, 3D touch-rotation interactions, and structured child-focused UX.
- Optimized interior lighting and scene setup for an ULCC architectural visualization project using baked lighting for photorealistic real-time walkthroughs.

KEY SHIPPED PROJECTS

- MetaGP Garage VR — First-person VR prototype to production; JWT auth, dynamic asset loading, 72+ FPS on Meta Quest 2
- Honeywell AR Training — Industrial petroleum plant simulation; debugged and optimized to 40%+ efficiency improvement
- GenRobotics VR — Real-time networked TCP video streaming on Oculus Quest, built end-to-end from scratch
- Serenity PathWays VR — Clinical cognitive training app with physics puzzles and progressive difficulty systems
- KidZania TVS Bike Studio / Mahindra Kiosk — Interactive kiosk experiences supporting 100+ daily sessions
- Auticare VR — ASD educational platform; interactive learning modules

EDUCATION

B.Tech, Computer Science & Engineering

College of Engineering Kottarakara, 2016 – 2020 | CGPA: 7.4/10

Arun Kumar A R
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Dear Hiring Team,

I'm writing to apply for the Senior Unity Developer (AR/VR) role at UST. I've spent the last five-plus years building Unity/C# applications across mobile, PC, and VR/XR platforms, with the majority of that work concentrated on immersive AR/VR delivery for enterprise clients — which lines up closely with what this role is asking for.

At Tiltlabs Consultancy Services, I built MetaGP Garage VR from prototype to production on Meta Quest 2 — JWT auth, Addressable Asset System, and a sustained 72+ FPS across all build iterations. I profiled and debugged Honeywell's AR training application, a petroleum plant simulation, improving training efficiency by over 40%. I also architected the GenRobotics VR experience, building real-time TCP socket-based video streaming end to end from scratch, and delivered Serenity PathWays VR, a clinical cognitive training app with physics-based puzzles and progressive difficulty scaling. Alongside the VR work, I've shipped interactive kiosk experiences for KidZania (TVS Bike Studio and Mahindra Lifespaces), including AWS S3 integration for a kiosk supporting 100+ daily sessions.

On the technical side, I work with SOLID principles and Unity-native design pattern implementation — Singleton, Observer, State Machine, Object Pool, and Factory — alongside URP/HDRP, Shader Graph, and Addressables. I also have hands-on experience with Unity 6, Mirror networking, and computer vision integration via OpenCV and Tesseract OCR in Unity — which I understand aligns with the computer vision specialization UST calls out as a plus. I'm comfortable optimizing performance across devices and collaborating closely with designers and backend developers to ship features end to end.

I'd welcome the chance to bring that AR/VR and Unity background to UST's team. Thank you for considering my application — I'm happy to share more detail on any of the projects above.

Best regards,

Arun Kumar A R